

CCNA 200-301 V2.0

12

Rapid PVST+ and the Protection Features

Part II — Switching and Network Access

EXAM TOPICS

2.5

LECTURER

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Where this chapter sits

Part I	Foundations	chapters 1–7
Part II	Switching and Network Access	chapters 8–14
Part III	IP Routing	chapters 15–19
Part IV	Network Services and Security	chapters 20–26
Part V	Wireless, Virtualisation and Operations	chapters 27–32
Part VI	Sitting the Exam	chapters 33–34

What you will be able to do

- **Explain** what a layer 2 loop does to a network, and why it is worse than any layer 3 failure.
- **Predict** the root bridge, root ports and designated ports for a given topology, and **influence** the outcome deliberately.
- **Configure** Rapid PVST+, root primary and secondary, PortFast, BPDU guard, root guard and loop guard.
- **Diagnose** an unexpected root bridge, an errdisabled port and a port that will not leave blocking.

Before we start

Chapter 3 for flooding — which is what makes a loop catastrophic — and Chapter 10 for trunks between switches.

Vocabulary for this chapter

BPDU

root bridge

bridge ID

bridge priority

root port

designated port

alternate port

path cost

Rapid PVST+

PortFast

BPDU guard

root guard

loop guard

errdisable

Each is defined where it first matters — not here.

Why a layer 2 loop is fatal



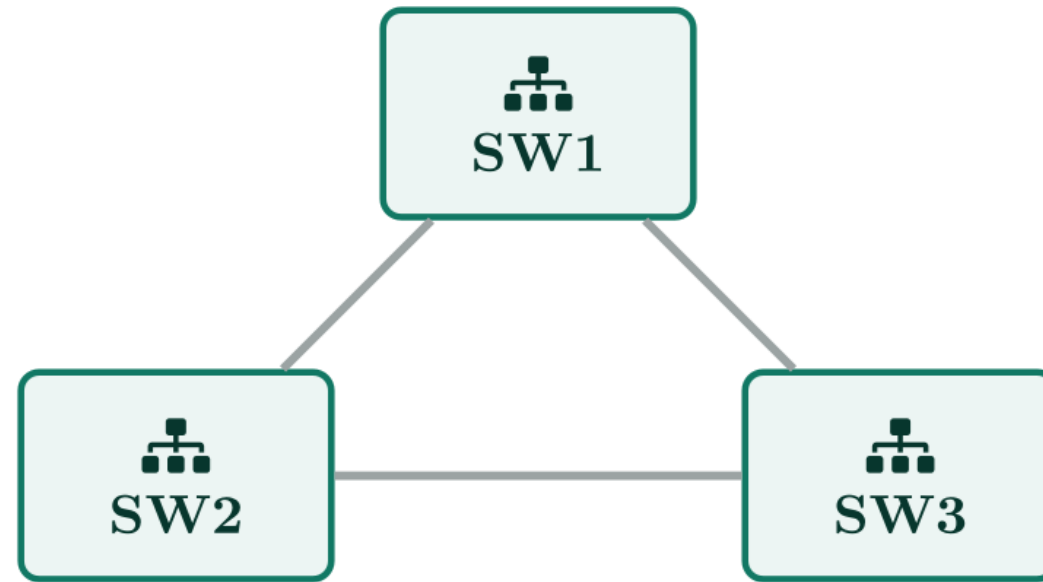
Three things happen at once, and none of them stops

An IP packet carries a TTL that decrements at each router, so a routing loop eventually discards the packet. **An Ethernet frame has no TTL.** A frame caught in a switching loop circulates until somebody unplugs something.

- **Broadcast storm.** A single broadcast is flooded, arrives back, and is flooded again — multiplying without limit until the links saturate.
- **MAC table instability.** The same source address arrives on two ports repeatedly, so the table is rewritten constantly and unicast forwarding becomes unreliable (see the flap message in Chapter 3).
- **Duplicate frame delivery.** Hosts receive multiple copies, which breaks protocols that assume they will not.

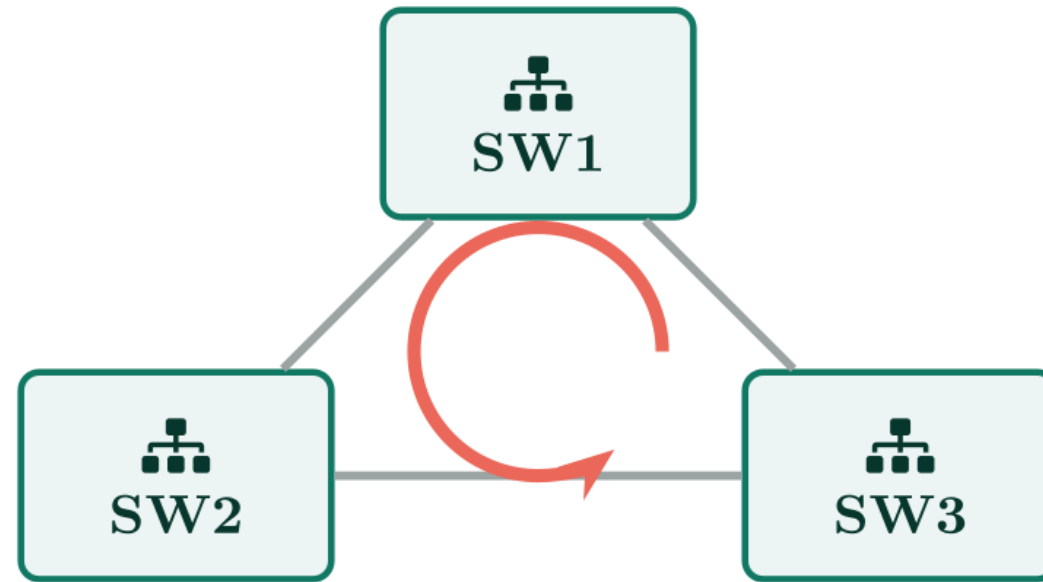
The network does not degrade gracefully; it stops, usually within seconds. This is why spanning tree exists and why it is on by default.

Why a layer 2 loop is fatal



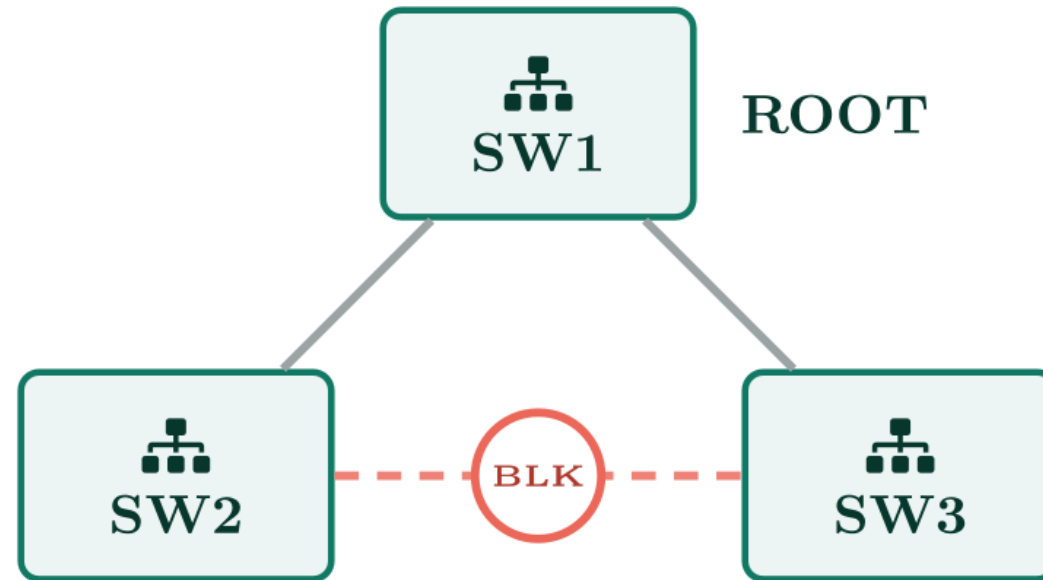
Three switches, three cables. The redundancy is deliberate.

Why a layer 2 loop is fatal



**One broadcast, and an Ethernet frame has no TTL
— it circulates for ever.**

Why a layer 2 loop is fatal



One port blocks. The cable is still there, ready if another link fails.

How the tree is chosen



Bridge ID and root election

Every switch has a **bridge ID**: a 4-bit priority, plus a 12-bit extended system ID carrying the VLAN number, plus its MAC address. The switch with the **lowest bridge ID wins** and becomes the **root bridge**.

Default priority is 32768. Because the extended system ID adds the VLAN number, the effective priority in VLAN 10 is $32768 + 10 = 32778$. Priorities are configured in steps of 4096.

Which of these three is the root, in VLAN 10?

Three switches, all left at defaults:

{@{}||@{}} SW1 & priority 32768 & MAC 0011.2233.0100 SW2 & priority 32768 & MAC 0011.2233.0050 SW3 & priority 32768 & MAC 0011.2233.0200

Step 1 — compare priority. All three are 32768, and all three are in VLAN 10, so all three have the same effective priority of $32768 + 10 = 32778$. No winner yet.

Step 2 — the tie-break is the MAC address, and lowest wins. Compare them left to right, as you would compare words alphabetically. The first eight digits are identical, so the decision comes down to the last four: 0100, 0050, 0200.

In hex, 0050 is the smallest of the three (Chapter 3 if the hex is not yet automatic — 0050 is 80 in decimal, 0100 is 256).

SW2 is the root bridge.

Now notice what just happened: **nobody chose this**. The winner is whichever switch happens to have the lowest MAC address, which correlates with nothing except manufacturing date. That is the point of the alert below.

Left alone, the oldest switch wins

If every switch keeps the default priority, the election is decided by the lowest MAC address — which usually means the oldest device in the building becomes the root of your network. That is almost never where you want traffic to converge. **Always set the root deliberately.**

Then every port takes a role

Once the root is known:

- Each non-root switch picks its **root port** — the one port with the lowest cumulative path cost to the root. It forwards.
- Each segment picks a **designated port** — the port on that segment closest to the root. It forwards.
- Everything else becomes an **alternate** (or backup) port and **blocks**. That is the loop being broken.

Ties are broken by lowest sender bridge ID, then lowest sender port ID.

Configuring Rapid PVST+



Rapid PVST+ with a deliberate root

IOS · configuration mode

```
! Run rapid per-VLAN spanning tree.  
spanning-tree mode rapid-pvst  
!  
! On the switch that SHOULD be root for these VLANs:  
spanning-tree vlan 10,20,30 root primary  
!  
! On the switch that should take over if it fails:  
! spanning-tree vlan 10,20,30 root secondary  
!  
! The macro above simply sets a priority. You can be explicit:  
! spanning-tree vlan 10 priority 4096
```

show spanning-tree vlan 10

SW1#

VLAN0010

Spanning tree enabled protocol rstp

Root ID Priority 4106
 Address 00d0.58a1.0200
 This bridge is the root
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 4106 (priority 4096 sys-id-ext 10)
 Address 00d0.58a1.0200

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----	----	---	-----	-----	-----
Gi0/1	Desg	FWD	4	128.1	P2p
Gi0/2	Desg	FWD	4	128.2	P2p

show spanning-tree vlan 10

```
SW2#
VLAN0010
Spanning tree enabled protocol rstp
Root ID      Priority    4106
              Address      00d0.58a1.0200
              Cost         4
              Port         1 (GigabitEthernet0/1)

Interface      Role Sts Cost      Prio.Nbr Type
-----
Gi0/1          Root FWD 4         128.1    P2p
Gi0/2          Altn BLK 4         128.2    P2p
```

The protection features



PortFast and BPDU guard belong together

PortFast on its own is a loop waiting to happen: it tells a port to forward immediately, so if somebody plugs a switch into it the loop is live before spanning tree can react. BPDU guard is the safety catch — the instant a BPDU arrives, the port is shut. Configure them as a pair, always.

Protection, per port and globally

IOS · configuration mode

```
! Per access port.
interface range GigabitEthernet0/5 - 20
    switchport mode access
    switchport access vlan 10
    spanning-tree portfast
    spanning-tree bpduguard enable
!
! Or globally, which applies to every access port automatically --
! far harder to forget on the next port somebody configures.
spanning-tree portfast default
spanning-tree portfast bpduguard default
!
! On a trunk facing a downstream switch that must never be root:
interface GigabitEthernet0/23
    spanning-tree guard root
!
! Loop guard, globally on all point-to-point links.
spanning-tree loopguard default
```

show interfaces status err-disabled

SW1#

Port	Name	Status	Reason	Err-disabled Vlans
Gi0/7		err-disabled	bpduguard	

Recovering an errdisabled port

Manually: `shutdown` then `no shutdown` on the interface — after removing whatever caused it. Automatically:

Automatic recovery is a convenience, not a fix. If a rogue switch is still plugged in the port will simply errdisable again every five minutes.

Common pitfalls

- **Leaving the root election to chance.** The oldest switch wins, and your traffic detours through the wiring closet nobody has visited in ten years.
- **PortFast without BPDU guard.** See above.
- **PortFast on a trunk to another switch.** It is for edge ports. There is a `portfast trunk` variant for specific cases, but the default answer is no.
- **Treating a blocking port as broken.** `Altn BLK` is spanning tree working.
- **Forgetting spanning tree is per VLAN.** A change to VLAN 10 does nothing for VLAN 20. Always name the VLANs.

After a new switch was added, traffic takes a strange path.

An access switch was installed in a spare closet. Since then, throughput between two core switches has halved and `show spanning-tree vlan 10` on the core reports that it is no longer the root.

After a new switch was added, traffic takes a strange path.

What would you check first — and what do you expect to see?

show spanning-tree vlan 10 | include Root ID|Priority|Address

CORE1#

Root ID	Priority	32778
	Address	0011.2233.4455
	Cost	19

What is the fault — and what does this output rule out?

Why it is doing that

Diagnosis. The root's priority is 32778 — the default of 32768+10 — and the MAC address is not the core's. The new access switch shipped with default priority and a lower MAC address than the core, so it won the election. Every VLAN's traffic now converges on a small access switch connected by a 100 Mbps link, which is where the cost of 19 comes from and why throughput halved.

And the lesson

Fix. Two steps, and do both. Set the intended root explicitly on the core: `spanning-tree vlan 10,20,30 root primary`, and the secondary on the other core switch. Then stop it happening again: apply `spanning-tree guard root` on the core ports facing access switches, so a switch advertising a superior BPDU is ignored rather than obeyed.

Steer the tree, then protect it

Topology. Three switches cabled in a triangle — deliberately looped — with PCs on two of them.

1. Cable the triangle with no spanning tree changes. Identify the root from `show spanning-tree` and explain *why* that switch won.
2. Record which port on which switch is blocking.
3. Set `SW1` as `root primary` and `SW2` as `root secondary` for VLAN 1. Re-check and record what moved.
4. Disconnect the link between `SW1` and `SW3`. Time how long traffic takes to recover and note which port changed role.
5. Enable PortFast and BPDU guard on an access port, then plug a switch into it. Record the log message and the port status, then recover the port.

Steer the tree, then protect it (continued)

6. Enable `spanning-tree guard root` on `SW1`'s port to `SW3`, then try to make `SW3` root by lowering its priority. Record what happens instead.
7. **Extension.** Time how long the network takes to recover from a link failure under `spanning-tree mode pvst`, then repeat under `rapid-pvst` and compare the two figures.

CHECKPOINT 1 of 3

Three switches have default priorities and MAC addresses ending `...0100`, `...0050` and `...0200`. Which is root, and why?

Discuss · then advance

Checkpoint 1 of 3

Three switches have default priorities and MAC addresses ending ...0100, ...0050 and ...0200. Which is root, and why?

The one ending ...0050. With equal priorities the lowest MAC address wins the tie-break, and 0050 is numerically lowest.

CHECKPOINT 2 of 3

A port shows Alt'n BLK. Is this a fault? Justify your answer.

Discuss · then advance

Checkpoint 2 of 3

A port shows Altn BLK. Is this a fault? Justify your answer.

No, it is correct and necessary. An alternate port blocking is spanning tree doing exactly its job — it has found a redundant path and is keeping it available without creating a loop.

CHECKPOINT 3 of 3

Why must PortFast and BPDU guard be configured together?

Discuss · then advance

Checkpoint 3 of 3

Why must PortFast and BPDU guard be configured together?

Because PortFast removes the delay that would otherwise let spanning tree detect a loop before the port forwards. If somebody plugs a switch into that port, a loop forms immediately. BPDU guard restores the safety by err-disabling the port the instant a BPDU appears — which is the only thing that should never arrive on an access port.

What to take away

- An Ethernet frame has no TTL, so a layer 2 loop never self-clears. A broadcast storm takes the network down in seconds.
- Lowest bridge ID becomes root; bridge ID is priority plus VLAN plus MAC. Set it deliberately or the oldest switch wins.
- Root port per switch, designated port per segment, everything else blocks.
- Rapid PVST+ runs one rapid instance per VLAN. Configure it, and remember every command is per VLAN.
- PortFast with BPDU guard on access ports; root guard facing downstream switches; loop guard on trunks. Errdisable is recovered by `shutdown/no shutdown` after removing the cause.

Where we got to

- An Ethernet frame has no TTL, so a layer 2 loop never self-clears. A broadcast storm takes the network down in seconds.
- Lowest bridge ID becomes root; bridge ID is priority plus VLAN plus MAC. Set it deliberately or the oldest switch wins.
- Root port per switch, designated port per segment, everything else blocks.
- Rapid PVST+ runs one rapid instance per VLAN. Configure it, and remember every command is per VLAN.

NEXT

Chapter 13 — CDP, LLDP and Documenting a Network